

Day 08: Outliers and Interquartile Range (IQR)

Understanding Extreme Values in Data

100 Days of Data Science

Why Outliers Matter in Data Science

In real-world datasets, some values are very different from the rest. These unusual values are called outliers.

Outliers can:

- Distort averages
- Mislead models
- Represent errors or rare events

Understanding outliers is essential before building ML models.

What is an Outlier?

An outlier is a data point that lies far away from the majority of values.

Example:

Student marks: 45, 48, 50, 52, 49, 51, 300

Here, 300 is an outlier.

Real-Life Example

Bank transaction amounts:

Most transactions: 500 – 5,000 One transaction: 5,00,000

This could be:

- Fraud
- Special case
- Data error

Outliers often contain important information.

Why We Should Not Ignore Outliers

- They can change the mean significantly
- They can affect ML predictions
- They may represent rare but important events

Example: Average income of 10 people becomes very high if one billionaire is included.

Quartiles

Quartiles divide data into four equal parts:

- Q1 – first quartile (25%)
- Q2 – median (50%)
- Q3 – third quartile (75%)

Interquartile Range (IQR)

IQR measures the spread of the middle 50% of data.

$$IQR = Q3 - Q1$$

It is not affected by extreme values.

Why IQR is Useful

- Detects outliers robustly
- Ignores extreme noise
- Works well for skewed data

Outlier Detection Using IQR

Lower Bound:

$$Q1 - 1.5 \times IQR$$

Upper Bound:

$$Q3 + 1.5 \times IQR$$

Any value outside this range is an outlier.

Simple Example

Suppose:

$$Q1 = 20 \quad Q3 = 40$$

$$IQR = 40 - 20 = 20$$

$$\text{Lower bound} = 20 - 30 = -10 \quad \text{Upper bound} = 40 + 30 = 70$$

Any value below -10 or above 70 is an outlier.

Outliers in Machine Learning

Outliers can:

- Break linear models
- Affect distance-based algorithms
- Bias predictions

Common algorithms affected:

- Linear Regression
- KNN
- K-means clustering

How Data Scientists Handle Outliers

- Remove them (if error)
- Cap them (winsorization)
- Transform data (log scale)
- Analyze separately

Common Mistakes

- Removing all outliers blindly
- Ignoring domain knowledge
- Using only mean-based methods

Final Takeaway

Outliers are not always bad.

They can:

- Reveal errors
- Indicate rare events
- Contain valuable insights

Smart Data Science understands outliers instead of blindly deleting them.